

Public datasets: <https://github.com/awesomedata/awesome-public-datasets>

<https://www.mygreatlearning.com/blog/database-projects/>

1. Online Retail Application Database:

Build a database to manage products, customers, orders, and payments for an online store. You can design modules for product listings, shopping carts, and checkout functions. This project allows you to explore functionalities like user accounts and secure transactions using appropriate security measures.

Entities: products, customers, orders, order Items, etc.

Bonus:

- Implement user accounts and secure login functionality.
- Explore integrating a shopping cart system with product selection and checkout functionalities.

2. Hospital Management System:

Develop a system to manage patient records, appointments, medical history, and staff information. This project can involve designing modules for patient registration, scheduling appointments, HIPAA compliant billing, and integrating with electronic health record (EHR) systems.

Entities: patients, doctors, appointments, bills, etc.

3. Electricity Bill Management System:

This project allows you to manage customer information, billing data (including electricity consumption rates), and payment history for electricity bills. You can practice writing queries to generate bills based on consumption and track payment statuses, potentially integrating with online payment gateways.

Entities: customers, billing rates, electricity usage, bills, etc.

- https://www.iqytechnicalcollege.com/Database_Management_System_Case_Studies%20Hospital.pdf
- <https://github.com/topics/dbms-project>

4. Bookstore Database:

<https://github.com/caesarmario/database-bookstore-case-study?tab=readme-ov-file>

In view of the growing population, the university is planning to establish an e-bookstore. The online store will facilitate the purchase of latest books and material of many genres. Your team is assigned the project to design a database system for online University e-Bookstore System.

- **Publishers** of books frequently send lists of latest books and materials to the e-bookstore manager. The bookstore manager compiles a list of needed books and sends an order to the publishers. The publisher supplies the ordered books to the university. The bookstore manager records the details of a new book, along with the number of the books that have arrived at the bookstore. An invoice is sent to the accounts department to be processed and payment made.
- **Customers**, who wish to purchase books online, need to initially register as members. Members will be able to view the book, read reviews and compare the online products with other similar articles.
- **Members** who wish to purchase can select their books into the website's shopping cart. The cart will show the summary of the selection and total cost to be paid. Once the payment is made, the customer will be able to print or save the receipts. The bookstore will send the books to the customers within 7 working days.

- The system should manage information about **books** in the bookstore, inventory, (registered) customers and books they have ordered. It should also store information about user opinions and book ratings.
- Users can also provide '**feedback**' for a book, as a score (1-10 which is 0= terrible, 10= masterpiece) along with optional short text. No changes are allowed; only one feedback per user per book is allowed.

5. College Data Management System:

Entities: students, courses, departments, instructors, etc.

<https://github.com/al-ghaly/Data-Management-Case-Study/blob/main/Project%20Documentations.pdf>

6. Hotel Booking management System:

<https://github.com/swetababu/HOTEL-MANAGEMENT-SYSTEM-DATABASE-PROJECT/blob/master/Documentation%20%26%20Presentation/Hotel%20Database%20Management%20System-SwetaGupta.pdf>

- A hotel is a hive of numerous operations such as front office, booking, and reservation, inventory, material management, quality management, security, energy management, housekeeping, CRM and more.
- A hotel can be apart of different Hotel Chains. A hotel chain may have multiple hotels. A hotel has multiple rooms and floors, and these rooms can be of different types. By room type, each room has the different price and description. A hotel has some employees to manage the services provided to customers. The customer can book the room either online or by cash payment at the hotel.
- The guest record is stored in hotel database which contains customer identity, his address, check in time, check out time, etc.

7. Cinema reservations system:

<https://github.com/enginooby-university/cinema-movie-ticket-booking-system-database-model/blob/master/Report%20issues.pdf>

- The **customers** table stores customers booking tickets. Only last name and address id fields are optional.
- The **addresses** table stores addresses of customers with each one assigned to one customer. Only id field is mandatory.
- The **films** table stores films which are shown in the cinema. The **length_min** is length of a film in minute. All fields are mandatory.
- The **auditorium** table stores data of all auditoriums in the cinema. The **no_seats** is number of seats in an auditorium. All fields are mandatory.
- The **screening** table stores data of all screenings which are identified by each unique pair of a film and an auditorium All fields are mandatory.
- The **seats** table stores all the seats in auditoriums with each seat assigned to strictly one auditorium. All fields are mandatory.
- The **bookings** table stores all the ticket reservations with each one assigned to one customer and one screening. If tickets were sold, the attribute **paid** would be set to 1 (true). All fields are mandatory.
- The **reserved_seats** table stores all the seats which are reserved by booking. Therefore, each one is assigned to one booking and one seat. All fields are mandatory.

8. Transport management system:

- **stations**: stores all stations data.
- **trains**: contains the trains information of the transport system.
- **passengers**: stores the personal information from the passengers that use the transport system.
- **routes**: saves the information on the train routes, relating the station to the trains.

- **travels:** stores the information about the trips made by the passengers, relating the routes with the passengers.
- **count_passengers:** It counts the passenger's amount every time that it's changed at the passenger's table.

9. Blood bank management system:

<https://github.com/phoenixx1/Blood-Bank-Management-DB-System/blob/master/PROJECT.pdf>

entities: donor, district, staff, blood_recipient, blood_sample, blood_processing_manager, hospital, disease_recognizer.

10. Car rental system:

https://github.com/naveenrajceg13/Car-Rental-System---Database-Project/blob/master/Project_Report.pdf